Lecture 18

Tuesday, 4 April 2023 10:33 AM

Tutorial on Thursday; Last week final project/presentation due

Ammonia synthesis as model reaction (Norskov group)

1) $N_2(g) + 2* \rightleftharpoons 2N^* \rightarrow \Delta E_1$,

2) $N^* + H^* \rightleftharpoons NH^* + * \rightarrow \Delta E_{NH} = E_{NH^*} - E_* - E_{NH(g)}$

How many independent descriptors do we need to describe trends in activities?

3) $NH^* + H^* \rightleftharpoons NH_2^* + * \rightarrow \Delta E_{NH_2}$

4) $NH_2^* + H^* \rightleftharpoons NH_3^* + 2*$

ΔE_H

5) $H_2(g) + 2* \rightleftharpoons 2H^*$

Scaling Relationships (real relationships that must be accounted for) → most accurate for species that bind using the same atom → H & N are linked via scaling relationship

$\Delta E_{NH} = 0.67\,\Delta E_N - 1.04$ (RPBE, eV)

$\Delta E_{NH_2} = 0.37\,\Delta E - 1.26$

Intercepts will change w/ functional (But slopes usually don't change)

$\Delta E_2 = E_{NH^*} + E_* - E_{H^*} - E_{N^*}$

$= -\Delta E_N - \Delta E_H + \Delta E_{NH} + \underbrace{(E_{NH} - E_N - E_H)_{gas}}_{\Delta E_{gas}}$; Can also use DFT to calculate energy of isolated species in gas phase

Plugging in ΔE_{NH} scaling relationship:

$\Delta E_2 = (0.67 - 1)\,\Delta E_N - 1.04 + \Delta E_{gas} - \Delta E_H$;

$\Delta E_H \approx -2.6$ eV (approximate this because H ads does not vary too much across different surfaces)

↳ Alternatives: ① Finding a scaling relation b/w ΔE_H and ΔE_N

② Keep ΔE_H as an independent descriptors

Recall:

$\Delta E_1 = (2E_N - E_{N_2})_{gas} + 2\Delta E_N$

↳ $\Delta E_N = \frac{\Delta E_1}{2} + \frac{(E_{N_2} - 2E_N)_{gas}}{2}$

$\therefore \Delta E_2 = -0.16\,\Delta E_1 - 0.16\underbrace{(E_{N_2} - 2E_N)_g}_{-9.47\text{eV}} - 1.04 - (-2.6) + \underbrace{(E_{NH} - E_N - E_H)_g}_{-3.76}$

$\Delta E_2 = -0.16\,\Delta E_1 - 0.69$

$\Delta E_6 = \underbrace{(E_{H_2} - 2E_H)_{gas}}_{4.58\text{eV} \rightarrow \text{constant for all catalysts}} + 2\Delta E_H = -0.62$ eV

$\Delta E_3 = -0.15\,\Delta E_1 - 0.22$; $\Delta E_4 = -0.19\,\Delta E_1 + 0.97$ → Slopes are similar due to the similarity of the elementary step.

Combining BEP and TST (Literature: J. Cat v192, p31, 2000)

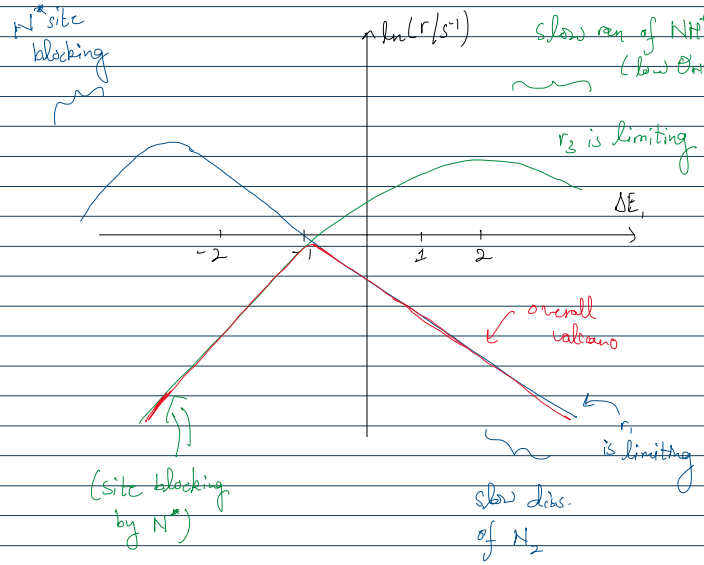
$$\rightarrow k_1 = \frac{1.2\times10^6}{\sqrt{T}}\left(\frac{\sqrt{K}}{bar}\right)\exp\underbrace{\left[\frac{-0.9\,\Delta E_1 + 1.34}{k_B T}\right]}_{\text{BEP relation}}$$

→ (Lit. J. Cat, V220, p273, 2003)

E_{a_3} (Ru step) = 1.35 eV

ΔE_3 (Ru step) = −0.83 eV

Assume, $\Delta E_{a_3} = \alpha_3 \Delta E_3 + \beta_3$; $\beta_3 = 1.51$ eV (0.9; $-0.15\Delta E - 0.32$) } Very approx. results



(Caution): Most paper often assume that step 1 is always rate-limiting

↓

(not always true)

Q: How can you have r_3 limiting & still be varying w/ ΔE_1 descriptor ????

Physical basis of scaling relationships

+ Usually results from bond-order conservation principle

Bond order conservation

eg: ΔE_{NH} vs. ΔE_N ; N can form 3 bonds ; BOC: When N-containing species adsorb the maintain valency by making an appropriate no. of bonds to surface

Thus, $\Delta E_N \approx 3 e_{N-M} + \text{const.} = (E_{N^*} - E_* - E_{N(g)})$

(e_{N-M}: average strength of N-metal bond (forms 3 bonds to the surface))

$\Delta E_{N-H} \approx 2 e_{N-M} + \text{const.}$; $e_{N-M} = \frac{1}{3}\Delta E_N \Rightarrow$ $\boxed{\Delta E_{NH} = \frac{2}{3}\Delta E_N + \text{const.}}$ ★

(only 2 bonds to the surface (since 1 is bonded to H))

This is very robust on metal surfaces, simple oxides (reducible), zeolites, 3-phase boundary

↓

if charge balanced

In general, $\boxed{\Delta E_{AB_n} = \frac{n_{max,A} - \chi}{n_{max,A}} \Delta E_A + \text{const.}}$ → first order approximation using just simple bond order analysis as done previously

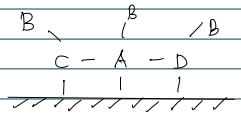
(AB_n: binds to surface; B is often H)

One data point will give const.: $\text{const.} = \Delta E_{AB_n/Pt} - \frac{n_{max} - \chi}{n_{max,A}} \Delta E_{A/Pt}$ Pt

$$\therefore \Delta E_{AB_x} - \Delta E_{AB_x/Pt} = \frac{n_{max} - x}{n_{max,A}} \left(\Delta E_A - \Delta E_{A/Pt} \right)$$

Slope is relatively insensitive to facet, but intercept is sensitive

Consider multidentate species:

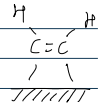


(eg)

Contribution from binding through C:

$$\Delta E_{CAB_2} - \Delta E_{CAB_2/Pt} = \frac{n_{max,} - n_{CAB_2}}{n_{max,C}} \left(\Delta E_C - \Delta E_{C/Pt} \right)$$

Assume contributions from C, A, D are independent → use linear sum } All bonds contribute a single bond order → This fails if we have multiple bonds b/w adsorbed species



$$\Delta E - \Delta E_{Pt} = \sum \frac{(n_{max,i} - n_i)}{n_{max,i}} \left(\Delta E_i - \Delta E_{i/Pt} \right)$$

But for eg if you have aromatics

In practice, higher order bonds are broken on surface, so scaling still holds (exceptions: Au, Ag, Cu) } Observation: Once in a while, species can form more than "one effective" bonds to the surface

Scaling is more accurate when correlating binding energies of species bound to same site geometries

↳ create separate scaling relations for diff site types.